

Advanced (Habanero)

Q1. Classify the number 5 as:

- (A) Natural and whole
- (B) Integer only
- (C) Whole only
- (D) Natural only
- (E) None of the above

Q2. What is the next even number after 14?

- (A) 12
- (B) 16
- (C) 18
- (D) 20
- (E) 22

Q3. Is the number -3 a:

- (A) Natural number
- (B) Whole number
- (C) Integer
- (D) Real number
- (E) All of the above

Q4. Fill in the pattern: 2, 5, 8, 11, 14, ...:

- (A) 15
- (B) 17
- (C) 18
- (D) 20
- (E) 22

Q5. Evaluate the expression $(-2) \cdot (-5)$:

- (A) -10
- (B) 10
- (C) -4
- (D) 4
- (E) 0

Q6. Simplify $3 + (-2)$:

- (A) 1
- (B) 5
- (C) -1
- (D) -5
- (E) 0

Q7. What is the value of x in the equation $x - 2 = 7$?

- (A) 5
- (B) 9
- (C) 10
- (D) 12
- (E) 15

Q8. Which property does the operation $a \cdot b = b \cdot a$ demonstrate?

- (A) Commutative
- (B) Associative
- (C) Distributive
- (D) Identity
- (E) None of the above

Open Ended Questions (FRQ)

Q9. Show that $(-a) \cdot (-b) = a \cdot b$ for any integers a and b . Use 5 lines or less.

Q10. Prove that if a is an even integer, then a^2 is also even. Use 5 lines or less.

Chef's Tips

- Remind students to show all work for FRQs
- Encourage students to use proper mathematical notation
- Emphasize the importance of checking answers for multiple-choice questions